

Normative Thermal Comfort Assessment

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Abstract

This paper reports on the development of a thermal comfort toolkit designed for the normative assessment of the indoor thermal environment of office facilities. It is one component of a larger building performance toolkit developed for the US General Services Administration. The thermal comfort part includes a set of unambiguously defined performance indicators (PI) based on the notion of objectively quantifiable performance measures. The set of indicators is founded on existing knowledge in biophysics and physiology, and constitutes a normative tool to assess the performance of buildings. The calculation of PIs is fully operational and web based. In the paper, the theoretical derivation of the indicators is given and their relevance in design scenarios is discussed.

Introduction

Providing a thermally comfortable environment is crucial for occupants' well-being and productivity.

Thermal discomfort can be caused by many factors beyond temperature, such as the layout of heat/cold supply in the space, varying space usage, occupants' activities, insufficient capacity or malfunctioning of mechanical systems, and other.

Thermal comfort assessments can be based on detailed simulations of the building design, or (if the building already exists) by on-site data gathering. Thermal comfort is derived by aggregating the simulated state information (air and mean radiation temperatures, humidity, and air velocity) and passing it through a thermal comfort model. The most used model for this purpose is Fanger's model [1]. This model subjects an observer to the current ambient conditions and depending on activity levels and clothing of the observer, a measure of thermal comfort at that time is generated. The measure is usually expressed as PMV or PPD. Fanger [1] established that if his model produces values for the PMV in the interval -0.5 to $+0.5$, then thermal comfort is adequate. The farther the PMV value outside this range, the higher the sense of discomfort. There has been a debate whether the Fanger model is adequate in all situations, as it ignores certain factors [2]. This, however, is not the topic of this paper. In what follows we have taken Fanger's model as the underlying comfort model, but used it in a method that produces thermal comfort assessments without having to resort to dynamic simulations. The latter typically require extensive time and efforts that go into modeling the room,

preparing the input model, checking the outcomes, and ultimately also doing a sensitivity or uncertainty analysis to investigate whether the outcomes are strongly influenced by certain modeling assumptions. An alternative method that does not require an extensive modeling and simulation effort is especially relevant for practitioners, in early design decisions, when design options need to be compared and judged on their thermal comfort impact. At that point one generally does not have all the information available that would warrant the effort to do a full dynamic comfort assessment. Often this type of detailed assessment is not possible or not desirable, the main reason being that a detailed assessment is laborious and requires many model assumptions that may bias or obscure the validity of the outcome. Moreover, and most importantly, such simulations produce detailed results but fail to capture and separate the impact of individual design factors on the resulting comfort realization. Our normative approach is an attempt to overcome all of this. In addition, it requires no modeling assumptions that go beyond what the design resolution may offer in the early design stages. This should not be confused with the notion of simplified simulations. Indeed, a simplified simulation is a “reduction” of more elaborate simulations that would be conducted with more refined inputs or more complicated models or both. The reduction does not focus on defining a standard experiment and a standard measurement method. But this is exactly what a normative method does. Our approach is hence not a simplified simulation but in fact a carefully constructed “virtual experiment” with a normatively defined quantification of an observable outcome. The design of the experiment is the key, as only experiments that reflect the inspection of a meaningful function of the design can be used. The second aspect of our approach is the way that the experiment is observed and translated into quantified outcomes. The aggregation of observed state variables is an intrinsic part of the virtual experiment, and thus the aggregated performance indicators represent true performance quantifiers. It should be noted that the current generation of simulation tools have not put much emphasis on standardizing aggregation of output states. This is, however, an essential part of our approach, and indeed of any normative approach.

The other major contribution of our approach is in its application potential in portfolio management of existing buildings. Thermal discomfort is still the number one complaint in office buildings [3], but it is hard to assess what investments in improvements will give the highest pay-off in comfort improvements. The proposed tool is capable of making that assessment, again without

resorting to detailed simulations. We will in fact show that in most office situations, when there is no individual (workstation) temperature control, that it is more the rule than the exception that more than 50% of the workers will experience discomfort. This confirms what most building managers already know, looking at their occupant complaints logs.

Obviously, discomfort cannot be separated from how people appreciate (subjectively) their own environment, and in many postoccupancy evaluations, the thermal comfort in existing spaces is assessed directly by questionnaire. In this paper, we ignore the subjective element and rely on Fanger’s model to generate an objective assessment purely based on the above mentioned four ambient variables and two observer properties mentioned above (activity levels and clothing). The main objective is to make this assessment not based on simulation but introduce the calculation methods of new measures that relate thermal comfort outcomes directly to design factors. These measures can be regarded as normative, and the objective of this paper is therefore to define a normative assessment of thermal comfort. The word “normative” literally means “conforming to or constituting a standard of measurement or value.” It is used in our context to mean that the measures are generated by a set of fixed calculations that could be eventually turned into a standard referenced by a building code. As long as this is not yet the case we use normative in the general sense, not in any specific code-mandatory or other more limited sense. In normative assessment, it is not the intention to derive an accurate reflection of reality but derive an objective indicator for the performance of the “comfort provision system” or in short, “comfort system.” The outcome of thermal comfort (objectively determined through Fanger’s model) is the result of the complex interplay of the building, its form and materialization, its control, its operation, its maintenance, its potential defects, its client behavior, etc. All of this taken together forms the comfort system. Note that this does not take any subjective psycho-social factors into account, as motivated before. One could go one step beyond this and look at the broader “human-organizational-environmental comfort system” composed of the physical environment and the organizational, human environment but this would take us into a domain where few, if any, objective metrics have been established [4]. An important aspect of a normative approach is that there is no essential difference between the assessment of the as-designed and the as-operated building. If both are captured by the same parameter values, then obviously the normative calculations will produce the same result. But, one will argue, in realized

buildings the difference between actually observed and objectively assessed performance is sometimes noteworthy. For example, when measured environmental data indicates that the indoor thermal environment has a poor comfort rating, it does not mean that this is synonymous with a low score of the objectively determined performance indicator. The two may only be indirectly linked because actual comfort at a given point in time may be influenced by extraneous causes rather than a low performing comfort system. It is not hard to imagine situations where discomfort occurs, e.g., when one worker with a dominant attitude controls the thermostat, or when remodeling of spaces over time has created poorly calibrated air supply to different zones, and good comfort in one space automatically produces discomfort in an adjacent space. The hardest part in deriving objective performance measures is dealing with the phenomena that lead to a lower comfort rating. For every phenomenon we must identify the building parameters and calculations that capture the essence of the phenomenon and measure (through calculations) its impact on discomfort. If all of the causal building inherent factors have been captured enough, the thermal comfort indicators separate the systemic comfort performance of the combination of building systems from any actual, dynamically realized local comfort in the building in use. It is in fact the underlying motivation of this paper to show that this can indeed be done.

The next sections present the normative measures and their calculations to assess the thermal comfort of office facilities. The approach is based on the broader notion of objectively quantifiable performance measures, which are embodied in a set of unambiguously defined performance indicators (PI).

Methods

This section introduces the concept of a virtual experiment as a formal quantification method of a PI. Figure 1 shows the basic notion of a performance “analysis function” (AF) as a mapping of input variables, environmental conditions and control variables, and system properties (p) to an observable state and consequently to a $PI(p)$ through a specified aggregation procedure.

Figure 1 can be used to understand the calculation of a PI for thermal comfort. Thermal comfort performance is delivered by the comfort system, the combined set of governing building systems, e.g., the heating, cooling, control, and enclosure systems. The calculation of the PI is

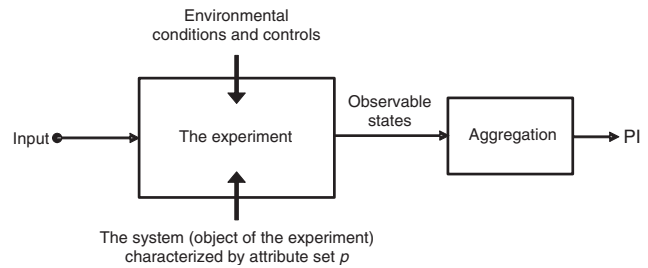


Fig. 1. Virtual experiment and aggregation that defines the PI.

based on the following experiment: a human is placed in a certain location in a given space of the building, which is subjected to a certain local climate and internal control regime. The experiment can be conducted virtually by performing a dynamic computer simulation. The experiment control variables are associated with the environmental and personal factors that lead to a certain level of thermal comfort such as thermostat control, ventilation actions (e.g., opening of windows), and observer properties, i.e., activity levels and clothing. Note that there is not one unique way to perform the aggregation over the output data of the experiment (the output data are composed of observable states of the experiment over time). In fact, alternative PIs for thermal comfort are in use today in building regulations, e.g., based on (1) number of hours per year that the room air temperature exceeds a certain comfort threshold or (2) the yearly PMV occurrence distribution [1]. Both PIs are derived from the same virtual experiment but use different metrics, in this case by performing a different aggregation method.

In our approach, we are neither interested in simulation nor real experiments. Rather a direct mapping from the set of system parameters (p) to some set of PI's is desired. This so-called normative approach rules out any bias by clearly stating unambiguous (normative) statements of how observable variables result in a particular quantification of the PI (e.g., a rating, score, or real number). Similar normative procedures are currently in use in other performance fields, e.g., the ST&M (Serviceability Tools and Methods) rating methods [5] and the sustainability rating method LEED [6]. All of the above provides the $PI(p)$ mapping, based on a set of rules that describe the experiment in precise terms to perform the following two steps without any ambiguity: (1) to perform experiment which results in *object states* and (2) to aggregate *object states* over time and space, resulting in the PI value.

As stated before, the virtual experiment could be performed by a full dynamic simulation. However, this is just one (full) experiment whereas we want to perform several experiments to single out the effect of specific

building parameters on different types of resulting discomfort. In addition, simulations are laborious and defeat the purpose of a quick evaluation or early design alternatives. A full simulation would introduce bias into the PI quantification as no simulation can be performed without introducing assumptions and simplifications, often dependent on the chosen type of simulation tool. The proposed normative calculation procedures, which are derived such that the results are indicative and objective indicators of a certain performance aspect, weed out this bias since our approach is not focused on “how” to calculate but “what” to calculate. The ‘how’ is in all cases first order (bio)physics based. The “what” is totally dependent on an agreed experiment. Simulation tools have no standardized experiment; rather they try to simulate a part of reality. Our approach does assume standardized experiments and is totally dependent on it. It should be noted that many more standardized virtual experiments could be thought of, in which case the number of PIs will increase proportionately. In that sense our approach is never complete, and should indeed go through a series of feedback loops with design teams. This is not unlike the development of standards related to testing of products in “real” experiment set-ups. If the standard body proposes a standard experiment, all parties can comment, amend, and ultimately accept or reject the standard. As we describe elsewhere in this paper that has not yet happened with our approach, except in small scale groups of clients that have used our approach in actual projects (such as within the GSA).

In this paper, a total of 11 normative performance indicators is introduced to indicate and “isolate” the discomfort attributable to air diffusion, asymmetric thermal radiation, draft, variation, zoning, and system’s response time. The indicators are offered through a web-hosted toolkit where users input building data through a web interface and the PIs are calculated instantly.

It should be noted that the thermal comfort performance toolkit is just one component of a larger building performance toolkit developed at Georgia Tech for the US Federal General Services Administration (GSA). The entire toolkit also includes indicators for energy, lighting, maintenance, and mold growth. For details about the complete toolkit, we refer to [7]. As an aside, it should be noted that GSA has been receptive to all PIs except those in the maintenance part [8]. It became clear that an organization like GSA could not readily accept the proposed standardized experiments that calculate the maintenance PIs. This is an example where more iterations with the portfolio management teams will be necessary.

The toolkit has been benchmarked on a set of GSA buildings in Georgia and on several buildings on the Georgia Tech campus. The following sections explain the thermal comfort indicators and their application on a Georgia Tech campus building. For want of space, all benchmark applications will not be addressed here, but Table 1 shows the list.

Eleven different PIs for thermal comfort were developed. Each indicator accounts for a specific phenomenon of potential discomfort, i.e., resulting from air diffusion, asymmetric radiation, cold draft, occupants’ variation, HVAC zoning, and the heating and cooling response time. The complete list is shown in Table 2. Each PI development is based on a scientifically proven method, which is then used to derive a certain performance measure, i.e. a measure that indicates the level of fulfillment of a desired capability of the comfort system, expressed in PPD. Each Performance Indicator is developed by isolating the effect of a specific type of environmental condition’s impact on the resulting PPD, if all other conditions are held neutral. The principles of the calculation methods are given below.

The air diffusion performance index (ADPI) (referred to as PI 1 in the paper) is a measure of how temperature

Table 1. Applications list

Building name	Location	Year built	Age	Floor area (m ²)
Sam Nunn	Atlanta	1996	8	148,822
Russell	Atlanta	1977	27	89,332
Peachtree Summit	Atlanta	1975	29	70,988
MLK	Atlanta	1934	70	30,389
Rome federal	Rome	1972	32	10,350
William Augustus	Macon	1908	96	7697
Stephens	Athens	1974	30	7621
Newnan Morgan	Newnan	1968	36	4680
Athens PO-CT	Athens	1941	63	3734
Federal Record Center	East Point	1945	59	33,210
Georgia Tech campus building (Klaus)	Atlanta	Under construction	N/A	17,960

variation, air mixing, and occurrence of drafts affect thermal comfort in the occupied zone of the space. The ADPI is defined as the percentage of test points where the effective draft temperature (θ) and air velocity (V) meet the desirable criteria as shown in Equation (1) [2] and can be regarded as an indicator of the percentage of people who are comfortable in office occupations. In other words, an ADPI of 100% means the most desirable condition. The ADPI is calculated using the Tables in ASHRAE [2] with information on the layout and specifications of the installed air diffusers, the volumetric airflow rate to the space, the characteristic length of the space, etc.

$$-1.7^{\circ}\text{C} \leq \theta \leq 1.1^{\circ}\text{C}, \quad V \leq 0.35 \text{ m} \cdot \text{s}^{-1} \quad (1)$$

To account for the asymmetric thermal radiation caused by hot/cold glazing surface, which is one of most common discomfort reasons in the perimeter zones of a space, three performance indicators (PI 2, PI 3, PI 4) were introduced. They are based on the PMV/PPD, estimated from the glazing surface temperatures and the corresponding mean radiant temperature (MRT) over an average year. For the sake of a normative calculation in order to draw out the effect of glazing temperature on thermal comfort without being disturbed by other factors, the following assumptions are made: (1) the room air temperature at a central location in the space is assumed to be the optimal room air temperature, defined as the desired temperature to achieve $\text{PMV}=0$, a thermally ideal condition with given occupant's office activity levels and clothing at that location in the space; (2) The air velocity is assumed $0.1 \text{ m} \cdot \text{s}^{-1}$ and the relative humidity is equal to 50%.

Based on the diurnal and seasonal hourly weather variations obtained from TMY2 (Typical Meteorological Years), the hourly glazing temperature is calculated by using a quasi steady-state calculation method with

information on the U -values of 48 different types of glazing systems provided in [2]. The corresponding mean radiant temperature is then estimated in accordance with ASHRAE standard 55-2004 [9], resulting in PI 2, PI 3, and PI 4 as follows:

$$\text{PI2} = \frac{\sum_{k=1}^n (100 - \text{PPD}_k)}{n} \quad (2)$$

$$\text{PI3} = \frac{\sum_{k=1}^n [\text{PPD}_k]^+}{n} \times 100 \quad (3)$$

$$\text{PI4} = \frac{\sum_{l=1}^p (100 - \text{PPD}_l)}{p} \quad (4)$$

where PI 2 is the average $(100 - \text{PPD})$, n is the number of data points (if the sampling time is 1 h over a year for nine office hours with five working days a week, n becomes $9 \times 5 \times 52 = 2340$), PI 3 is the percentage of hours where the PPD is in comfort range ($< 10\%$), $[\text{PPD}_k]^+$ is set to equal to 1 if $\text{PPD}_k \leq 10\%$, PI 4 is the average of $(100 - \text{PPD})$ where $\text{PPD} \geq 10\%$, p is the number of occurrence where $\text{PPD} \geq 10\%$. If $p = 0$, PI 4 is set to 100%.

Regarding the cold draft, as distinguished from the radiant effects addressed above, due to cold glazing surface, three performance indicators (PI 5, PI 6, PI 7) are introduced, based on the percentage of the population feeling draft when exposed to a given mean air velocity. With the calculated hourly glazing temperature used for PI 2, the induced draft, the maximum air velocity [$U_{\max}(x)$ ($\text{m} \cdot \text{s}^{-1}$)] is calculated [10]:

$$U_{\max}(x) = \begin{cases} 0.055\sqrt{h\Delta t} & \text{if } x < 0.4 \\ 0.095 \frac{\sqrt{h\Delta t}}{x + 1.32} & \text{if } 0.4 \leq x \leq 2.0 \\ 0.028\sqrt{h\Delta t} & \text{if } x > 2.0 \end{cases} \quad (5)$$

Table 2. Performance indicators

Function	PI	Meaning	Calculated by
Air diffusion	PI 1	% of occupants in comfort	[2]
Asymmetric thermal radiation due to hot/cold glazing	PI 2	Hourly average $(100 - \text{PPD})$ during office hours over an year	[2,9]
Cold draft caused by glazing	PI 3	Percentage of hours where the PPD is in comfort range ($\text{PPD} \leq 10\%$)	[10,11]
	PI 4	Average of $(100 - \text{PPD})$ where PPD is not in comfort range ($\text{PPD} \geq 10\%$)	
	PI 5	hourly average $(100 - \text{PD})$ during office hours over an year	
	PI 6	Percentage of hours where the PD is in comfort range ($\text{PD} \leq 10\%$)	
Occupants' variation	PI 7	Average of $(100 - \text{PD})$ where PD is not in comfort range ($\text{PD} \geq 10\%$)	[2]
	PI 8	Average $(100 - \text{PPD})$ of workers in different activities and clothing	
	PI 9	Airflow rate variation in different rooms within a single thermostat zone	
System's capacity and response time	PI 10	The minutes required to increase the zone temperature by 1°C under the peak heating load	[12,14]
	PI 11	The minutes required to decrease the zone temperature by 1°C under the peak cooling load	

where, h is the height of the vertical glazing surface (m), Δt is the temperature difference ($^{\circ}\text{C}$) between the cooled surface and the reference in the occupied space, x is the distance (m) from the cold vertical glazing surface. Based on the calculated maximum air velocity, the percent dissatisfied (PD) due to draft is calculated using the model in [11] as follows:

$$\text{PD} = 13800 \left[\left(\frac{\bar{v} - 0.04}{t_a - 13.7} + 0.0293 \right)^2 - 0.000857 \right] (\%) \quad (6)$$

Where t_a is the air temperature and $\bar{v}$ is the mean air velocity, which can be estimated in relation to $U_{\max}$ [11]. Finally, in the same manner, the PIs 5–7 are calculated.

$$\text{PI 5} = \frac{\sum_{k=1}^n (100 - \text{PD}_k)}{n} \quad (7)$$

$$\text{PI 6} = \frac{\sum_{k=1}^n [\text{PD}_k]^+}{n} \times 100 \quad (8)$$

$$\text{PI 7} = \frac{\sum_{l=1}^p (100 - \text{PD}_l)}{p} \quad (9)$$

where, PI 5 is the average $(100 - \text{PD})$, PI 6 is the percentage of hours where the PD is in comfort range ($< 10\%$), $[\text{PD}_k]^+$ is set to equal to 1 if $\text{PD}_k \leq 10\%$, PI 7 is the average of PD where $\text{PD} \geq 10\%$, p is the number of occurrence where $\text{PD} \geq 10\%$. If $p = 0$, PI 7 is set to 100%. It should be noted that the PI 5, PI 6, and PI 7 can apply only to persons wearing normal indoor clothing and performing light, mainly sedentary office work. Persons with higher activity levels are not so sensitive to draft [15].

To account for the occupants' variation in office activity levels and clothing, PI 8 is introduced, defined as the average $(100 - \text{PPD})$ of different workers under the condition that the HVAC system provides an ideal thermal environment for the *design person* in a space. In other words, this indicator represents the difference in thermal comfort level between the design person that the HVAC system aims to satisfy and workers that have office activities and clothing different from the design person.

This indicator is useful to predict the variation in thermal comfort solely as a result of natural variations in activity levels (MET) and clothing (CLO) in the same space. This will interest employers especially if they have great diversity of activities and dress code among their workers.

To evaluate HVAC zoning problems, a conventional technique is used. For zones to work with single-point thermostat control, the proportional capacity requirement must not change significantly for a variety of internal and external load variations [12]. Thus, PI 9 is introduced to reflect the load variation of different spaces in the single control zone. If the airflow at peak loads of the different spaces in the single control zone varies more than 10–15% throughout the year, additional zoning is required [12]. The peak heating and cooling load calculations follow the Radiant Time Series method (RTS method) in [2]. Table 3 shows the exemplary calculation of PI 9 for Rooms 1, 2, and 3.

Finally, to address the system's capacity and response time, PIs 10 and 11 are introduced. With information on the size and efficiency of the chillers and boilers and the peak heating and cooling load, the system's capacity is investigated (satisfactory/unsatisfactory) and the response time (T), defined as the minutes required to increase/decrease the zone temperature by 1°C under the peak load, is calculated as follows:

$$T = \frac{MC_p}{Q_{\text{mech}} - Q_{\text{load}}} \times \frac{1}{60} \quad (10)$$

where, MC_p is a lumped heat capacity of a building as a multiplication of the mass (M) and specific heat (C_p), Q_{mech} is the total energy in W that is generated by the installed chillers or boilers and then delivered to the space, and Q_{load} is the peak heating and cooling load in W that has to be removed from the space, calculated by the aforementioned RTS method. It should be noted that Equation (10) is derived in a lumped steady-state fashion and, thus, does not reflect the reality very accurately compared to 3D full-blown dynamic simulations.

Table 3. Airflow requirement per space

Room	July (cooling)		January (heating)		Calculated PI 9
	Airflow (CMH)	Percent of total airflow	Airflow (CMH)	Percent of total airflow	
Room #1	3175	62.6%	2358	63.8%	$100 - 62.6 - 63.8 = 98.8\%$
Room #2	1026	20.3%	644	17.5%	$100 - 20.3 - 17.5 = 97.2\%$
Room #3	868	17.1%	688	18.7%	$100 - 17.1 - 18.7 = 98.5\%$
Total	5069	100%	3690	100%	

This also means that the calculated response time in Equation (10) does not capture the nuances of the load and the capacity in a specific case. Rather it is a characteristic time constant which is a good indicator of the dynamic response performance. This is another reflection of the normative approach that is advocated here. But this also creates a problem: it is hard to judge what the range is for the time constant to be sufficient. This requires that one knows the relationship between the computed response time and any resulting discomfort, i.e., resulting from a certain response time of the system. Such a relationship could be established by simulating different loading, set-back, and response scenarios in a random set of buildings. It should be possible to use these simulation outcomes to identify and calibrate the dominant parameters and develop a normative relationship. This work is planned as a follow-up.

The implementation of the thermal comfort toolkit is done using the Microsoft ASP.NET web application programming. The web-enabled execution of the toolkit provides the following advantages: (1) easy data entry and storage; (2) ubiquitous access to the use of the toolkit; (3) data sharing with other electronic repositories; (4) user friendliness. Figure 2 shows an example of a data entry screen in the web browser mode.

To calibrate/validate the toolkit and investigate its applicability, usability and adequateness, a medium scale benchmark was conducted on ten selected GSA facilities (Table 1). The benchmarking process consisted of three stages; (1) site visits (field survey and data collection), (2) data entry to the toolkit, and (3) calculation of PIs and result analysis. Through the benchmarking process, the current version of the toolkit was validated with metered, measured, and gathered data. More details on the

results of the medium benchmarking will be reported elsewhere.

Results

The toolkit is a vital support instrument in building design, as it shows at a very early stage the consequences of system properties on a selected performance aspect. A set of well-chosen performance indicators express the causal relationships between design decision and resulting performance. The performance indicators are computed by a set of normative formulas, executed in a web-hosted performance toolkit.

The application of the toolkit in a particular building is shown below. It concerns a new three-storey Georgia Tech campus building (the Klaus building), which was completed in 2006 and occupied in early 2007. The floor area, exclusive of the parking deck is 17,960 m² and the expected number of occupants is 2450. The HVAC system uses VAV with reheat.

For the calculation of PIs, it should be noted that PIs 1–9 investigate the local thermal comfort while PIs 10 and 11 assess the HVAC system’s capacity over the entire building. The calculation of the PIs 1–9 was made for two typical office spaces, Rooms 2362 and 2142 as shown in Figure 3.

Table 4 shows the calculated PIs. The calculated PI 1 was 73.6% in Room 2362 (12” neck, 4’ long, 1’ wide slot diffuser) and 84.0% in Room 2142 (2’ x 2’ perforated ceiling diffuser) respectively. In order for Room 2362 to achieve higher PI 1, the HVAC designer could have considered the location and type of the diffuser.

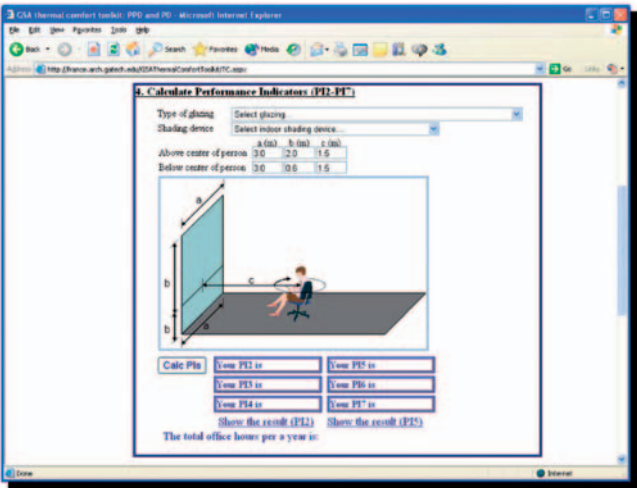


Fig. 2. Thermal comfort PI entry example: calculation screen.



Fig. 3. The second floor plan and selected two rooms for PIs 1–9.

Table 4. Calculated thermal comfort PIs

Function	PIs	Room 2362	Room 2142
Air diffusion	PI 1	73.6%	84.0%
Asymmetric thermal radiation due to hot/cold glazing	PI 2	95.0%	94.8%
	PI 3	100%	100%
	PI 4	100%	100%
Cold draft caused by glazing	PI 5	99.6%	99.3%
	PI 6	98.3%	96.7%
	PI 7	86.8%	83.5%
Occupants' variation	PI 8	L. I.*	L. I.*
HVAC Zoning	PI 9	100%	100%
System's capacity	PI 10	5.5 min	
and response time	PI 11	Lack of capacity	

*L.I.: Lack of Information.

The theoretical maximum ADPI with ceiling slot diffuser is 92% when appropriately selected and located. In the toolkit, the calculation of the ADPI-based PI 1 is quick and easy in any standard web browser without tedious calculation process, so any diffuser alternative can be very quickly checked for improved performance.

PIs 2–4 represent asymmetric thermal radiation due to hot/cold glazing. The hourly average over an entire year is 95.0 and 94.8%, respectively. There is no occurrence where $PPD \geq 10\%$ over a year. This means that asymmetric thermal radiation in both rooms is negligible since the theoretical minimum PPD is 5.0% when $PMV = 0$. It should be noted that at this point there was no definite information about the location of the workstations in the space. If a workstation is going to be located close to windows, the resulting PIs 2–4 will change accordingly. In fact, the proposed PIs can be used to verify how different layouts of workstation location in different spaces will affect the normative comfort performance. This can prove to be a vital instrument for space remodeling or space re-allocation decisions confronting facility managers on a daily basis.

In case that the calculated PIs 2–4 are less than a threshold value suggested by any of the building stakeholders (occupant, portfolio manager, designer), an alternative could be evaluated with ease by changing location, size or type of windows from the toolkit menu (Room 2362 is oriented to the east and uses 8'4" (H) $\times$ 11'6" (W) double glazing, while Room 2142 is to the north-west and uses 7'4" (H) $\times$ 10'0" (W) double glazing). This is a clear example where the quick and effortless assessment of normative performance delivers design support with a turn around time (minutes) that

corresponds to the demands of current design office environments.

Regarding the cold draft due to cold glazing surface, PIs 5–7 are calculated. Compared to PIs 2–4, there is higher possibility of discomfort due to the draft in the considered spaces.

It is important to note that when PIs address the effect of unrelated aspects of the system, the resulting PIs are "orthogonal." This means that the overall resulting performance is the product of the individual performances (as in this case they are all based on PPD). This orthogonality is similar to the approach encountered in some of the existing physical testing standards. For example, safety glass must (amongst other) provide safety against flying objects and before a product can only enter the market if it passes certain standard tests. As we know that the actual loading conditions are infinitely varied, we could try to mimic hurricane winds, fires, and other incidents. We could do this by throwing different objects with a certain velocity at a test set-up in a test facility. Not only would this be an expensive experiment but it is extremely hard to verify that an experiment in Denmark corresponds with the experimental set-up in Japan. Consequently it would be very hard to implement a unique and honest admission test for new products. (Nevertheless, to elucidate the point that was made earlier, this is what happens when full blown thermal comfort simulations are performed on the same building by two different consultants!) To explain this in greater detail we will discuss a standard testing method of the safety of glass partitions (ASTM F1915-05 [16]). For want of space, the explanation below is hugely simplified, but it serves to elucidate the point. The standard safety test experiment could be conducted according to the following protocol, describing a series of experiments:

Exp. 1: Using specified blunt and sharp impactors, begin a series of strikes against the center of the glass panel for the number of required impacts, first with the blunt impactor followed by the sharp impactor. The strikes shall be uniformly made with 9 s intervals. Record the number of strikes required to produce the first penetration of the panel (PI1), and the number of strikes required to produce an opening large enough to pass a 5" $\times$ 8" $\times$ 8" rigid rectangular box with no more than 10 lbf (PI2).

Exp. 2: Using a specified apparatus, start the propane torch and begin measuring the elapsed time. Start the series of strikes at the center of the glass panel after 9 s. Continue impacting at 9 s intervals until the required number of impacts is reached. Record the number of strikes required to produce the first penetration of the

panel (PI3), and the number of strikes along with the duration of the flame exposure required to produce an opening large enough to pass a 5" × 8" × 8" rigid rectangular box with no more than 10 lbf (PI4).

These are orthogonal experiments: they use the same object but apply a different experiment, and measure the time until a certain defined limit state response to the experiment occurs. The advantage is that they are repeatable and controllable. And each manufactured new product gets a fair transparent testing.

To apply the PIs in practice, of course we need to study what potential conditions may occur in a certain location and translate this in acceptable values of PI1, PI2, PI3, etc. This has everything to do with the assessment of acceptable risks over the life time of the facility. This assessment is far from easy in the case of glass failure, just as it is far from easy for a client to decide what is acceptable for our thermal comfort indicators. It is not hard to imagine that a downtown lawyer's office will have different expectations than a second hand tire sales office. In both the glass safety example as in thermal comfort case the key to manage different expectations for different aspects of safety or comfort, is the introduction of a set of orthogonal PIs, orthogonal in the sense that the experiments and/or their measurement methods are mutually independent.

It is interesting to investigate the product $PI\ 1 \times PI\ 2 \times PI\ 5$ which signifies the combined effect of air diffusion, asymmetric thermal radiation and cold draft. The result is as follows:

- Room 2362: $PI\ 1 \times PI\ 2 \times PI\ 5 = 69.6\%$
- Room 2142: $PI\ 1 \times PI\ 2 \times PI\ 5 = 79.1\%$

It is significant that even in high performing offices the product easily can reach the 60% range, indicating that there is a good chance that 40% of the office population may complain about discomfort! At the time of writing the building was not yet occupied, hence PI 8 cannot be calculated due to lack of information. As to HVAC zoning, Rooms 2362 and 2142 are grouped with two adjacent rooms having the identical heating and cooling load profiles. Hence, PI 9 becomes 100%.

Tables 5 and 6 show the calculated PIs 10 and 11, respectively. As shown in Table 6, the cooling system's capacity is slightly less than the maximum cooling load, leading to the incapability to decrease the room temperature in the maximum load condition. But, it should be noted that this occurs only under the worst case which occur only for a short time.

Table 5. PI 10 calculation (the minutes required to increase the zone temperature by 1°C under the peak heating load)

Heating system (kW)	Max heating load (kW)	Max heating occurrence	Ratio	PI 10
2145	1442	7 a.m. (Jan 21st)	149%	5.5 min

Table 6. PI 11 calculation (the minutes required to decrease the zone temperature by 1°C under the peak cooling load)

Cooling system (kW)	Max cooling load (kW)	Max cooling occurrence	Ratio	PI 11
1942	2119	3 p.m. (July 21st)	95%	Lack of capacity

Conclusions

This paper introduces a set of normative, objective thermal comfort performance indicators. They are harnessed in a web-based thermal comfort toolkit software. The toolkit is ideally suited to assess floor arrangements, air diffusion system, HVAC layout, and glazing system, in the early design stage. To this end, a set of mutually independent experiments are defined, not unlike the protocol used in physical testing standards such as glass safety experiments. The approach calculates different aspects comfort performance of a proposed building materialization, use and layout in separate experiments. In each separate experiment the thermal comfort system is studied in its actual design context.

The toolkit provides the standardized and normalized tests that help to make early rational design decisions. In addition, the most common complaint in buildings (too hot or too cold) can be readily diagnosed with the use of the toolkit.

The construction world is slowly moving toward performance-based design but this will only happen if the research community can come up with rich sets of normatively defined PIs. This is not an easy task. Haphazardly defined PIs may not make sense and will not support the specification of objectified expectations or support design decisions. This paper attempts address the underlying issues through the introduction of PIs in the relevant thermal comfort area.

As part of this ongoing research, these follow-ups are being conducted:

- Extended benchmarking will be conducted to cover a broader range of building types and tenants, relying on self-deployment of the toolkit by design firms and

consultants. Until now the benchmark has been limited to office buildings in the Georgia. Extension to other climate zones and specialty type buildings will test the broader applicability.

- The next step is to migrate the approach into the management process of large building portfolios, such as is the case on campuses, within GSA, and for large corporate owners. In those cases, the toolkit outcomes need to be related to company performance targets. For instance, in case of new design an owner organization can decide to set a company target for every PI and potentially use those targets in a performance contract with the A/E firms. Before such targets can be defined,

an owner needs to engage in “norming” exercise to identify reasonable and desirable targets. For existing buildings, the PI outcomes can be used to diagnose buildings with a consistently high number of comfort complaints. In many cases, the performance assessment will confirm that the comfort performance is systemically low and not the result of an undiagnosed pathology of the space.

In a benchmark, the use of the indicators showed that the combined effect of all discomforts can easily add up to 40–50%. This underlines why discomfort is still complaint number one, even in well-designed office buildings.

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